

DNS (DOMAIN NAME SYSTEM)

Internet Ka Asan Phonebook (With Visual Pictures) | Prepared by Syed Ghous

1. DNS Kya Hai? (Domain Name System)

DNS ka asal kaam Domain Names (jaise google.com) ko IP addresses (jaise 142.250.190.46) mein convert karna hai.

Computer boht seedha saadha hota hai, use English ke naam samajh nahi aate, woh sirf **Numbers (IP Address)** ko janta hai. Lekin hum insano ke liye numbers yaad rakhna boht mushkil hai! Sochein agar aap ko Facebook, YouTube, aur WhatsApp chalane ke liye har ek ka alag number yaad rakhna padta to dimagh thak jata. Is liye hum naam istemal karte hain, aur computer ke liye DNS use number mein badal deta hai.

2. Real Life Misaal (Mobile Ki Contacts List)

Isay bilkul aam zindagi ki misal se samajhte hain:

AAP KE MOBILE KA PHONEBOOK:

Sochein aap ke mobile mein aap ke dost ka number save hai. Jab aap ne use call karni hoti hai, to aap mobile khol kar uska naam dhoondte hain, maslan: "**Dost**". Aap "Dost" par click karte hain to call lag jati hai.

Lekin kya mobile "Dost" ke naam par call karta hai? **Nahi!** Mobile asal mein us number par call karta hai jo us naam ke pechy save hota hai (jaise 0300-1234567).

Internet par **DNS** bilkul mobile ki contacts list ya phonebook ki tarah kaam karta hai. Aap computer ko naam batate hain (google.com), DNS apni diary khol kar uska number (IP address) nikalta hai aur computer ko de deta hai!

3. DNS Kaam Kaise Karta Hai? (Visual Picture Flow Chart)

Jab aap mobile ya laptop par koi website likhte hain, to background mein yeh char steps hote hain:

DNS Flow Diagram by Syed Ghous

(Yeh poora kaam ek second ke 100th hisse mein ho jata hai, is liye hamein pata bhi nahi chalta!)

4. Cybersecurity Point of View (Hacker Kaise Dhoka Deta Hai?)

Cybersecurity ke student ke tor par aap ko pata hona chahiye k DNS par hacker kaise attack karte hain:

DNS Spoofing / Cache Poisoning:

Sochein agar koi aap ki mobile ki contact list mein se apni bank ke manager ka number badal kar apna number likh de? Jab aap manager ko call karein gy, to call hacker ko chali jaye gi!

Hacker internet par bhi yahi karta hai. Woh DNS server ke andar ja kar website ka asli IP address badal kar apni **Bakwas/Nakli (Fake) Website ka IP address** likh deta hai. User likhta google.com hai, lekin DNS use hacker ki kharab website par bhej deta hai, jahan uska password chori ho jata hai.

Syed Ghous Ki Expert Security Tip:

"DNS ko is dhoke se bachane ke liye hamesha **DNSSEC (DNS Security Extensions)** ka istemal kiya jata hai. Yeh DNS ke records par digital sign laga deta hai takeh koi hacker is mein nakli IP address daal kar bacho ko ya users ko dhoka na de sakay!"